

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Either Gas or radiation force/pressure and gravitational force / gravity (1) are balanced / equal and opposite / are in equilibrium (1) Or Outward force and inward force (1) are balanced / equal (1)	2			2		
		(ii)		Light from Spica will take 261 years to reach Earth (or v.v.) / this is the distance light travels in 261 years	1			1		
	(b)			Indicative content: Properties 1. Remainder of cycle: red giant → white dwarf... 2. Becomes larger, lower temperature / redder and brighter → smaller, higher temperature / whiter and dimmer. Reactions 3. Hydrogen fusion stops in centre of Sun and helium fusion starts – linked to red giant. 4. Helium fusion stops – no more fusion – linked to white dwarf. Forces 5. Red giant: Gas and radiation pressure in the core increases – linked to helium fusion – giving unbalanced forces and star expands 6. White dwarf: Gas and radiation pressure decreases when fusion stops causing star to shrink 7. End of red giant phase – outward forces cause outer layers to be blown away, contributing to a <i>planetary nebula</i> , leaving hotter core Other relevant points could be made – contributing to reactions or forces, e.g. When H-fusion stops the core shrinks and heats up until a new balance of forces is established	3	3		6		

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				5 – 6 marks At least 5 points covered in all three aspects <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks At least three points covered in two of the three aspects <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Up to two points from any aspect. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 3 total	6	3	0	9	0	0